TASK 3 – DEEP-DIVE ANALYSIS

Deep-Dive Analysis & Interactive Dashboard | ApexPlanet Data Analytics Internship

1. Objective

The objective of Task 3 was to perform a deep-dive analysis of the sales dataset, formally define core Key Performance Indicators (KPIs), and build an interactive dashboard that allows users to explore the selected business area. The selected deep-dive area for this task is customer segmentation based on age.

2. Dataset Overview

The analysis uses the cleaned sales dataset prepared during the earlier internship work. The dataset contains transaction-level sales information including order details, customer attributes, product and category information, quantity, unit price, city, gender, age, and total sales.

3. Core KPIs

KPI	Formula	Value	Business Rationale
Total Sales	SUM(Total_Sales)	■139.40M	Measures overall sales generated across the analyzed transactions.
Total Transactions	COUNT(Order_ID)	1,000	Measures the number of recorded transactions and overall purchase activity.
Total Quantity Sold	SUM(Quantity)	~5K	Measures the total number of units sold and overall sales volume.
Average Quantity / Order	Total Quantity ÷ Total Transactions	5.44	Shows the average number of units purchased per transaction.
Average Order Value	Total Sales ÷ Total Transactions	■139.40K	Shows the average sales value generated per transaction.

4. Deep-Dive Analysis – Customer Age Segmentation

Customer segmentation was selected as the deep-dive business area. Customers were grouped into three age segments using business rules:

Age Segment	Business Rule
Young	Age < 30
Adult	30 ≤ Age < 60
Senior	Age ≥ 60

5. Dashboard Analysis

Sales by Customer Age Segment: The dashboard shows that the Adult segment contributes the highest total sales, followed by the Senior and Young segments.

Quantity Sold by Age Segment: The Adult segment also records the highest quantity sold, followed by Senior and Young.

Category Sales by Age Segment: The category-level visualization shows that the contribution of the three age segments varies across categories such as Electronics, Education, Grocery, Furniture, and Fashion.

Interactive filtering: The dashboard includes Age Segment, Gender, City, and Category slicers so users can explore the same KPIs and visualizations under different selections.

6. Interactive Filter Observations

Filter Selection	Observed Total Sales
Female customers	■66.94M
Male customers	■72.46M
Hyderabad	■9.46M

These values are descriptive observations from applying the dashboard filters to the analyzed dataset. They do not by themselves establish causal relationships.

7. Key Findings

1. Adult customers contribute the highest sales among the three age segments.

2. Adult customers also account for the highest quantity sold.

3. Category contribution varies across the different age segments.

4. Female customer sales total ■66.94M, while male customer sales total ■72.46M in the analyzed dataset.

5. Hyderabad records ■9.46M in sales when selected through the city filter.

8. Business Implications

The segmentation analysis provides a structured way to understand how sales activity is distributed across customer age groups. The Adult segment represents the largest observed contribution to both sales and quantity in this dataset. The category-level breakdown can be used for further investigation of category preferences across customer segments. The gender and city slicers provide additional ways to explore demographic and geographic differences in the observed sales data.

9. Tools Used

Power BI: dashboard development, interactive filtering, KPI visualization and deep-dive analysis.
DAX: KPI and calculated-field creation.
Excel/CSV: cleaned sales dataset and supporting data preparation.

10. Conclusion

Task 3 demonstrates the use of KPI definition, business-rule-based customer segmentation, data visualization, and interactive dashboard design. The completed Power BI dashboard brings these analyses together and enables users to explore sales performance across age, gender, city, and category dimensions.

Submission Deliverables: Deep-dive report, GitHub repository, interactive dashboard/live dashboard link, and a 5–7 minute LinkedIn dashboard demonstration.